Introduction to DRAM System

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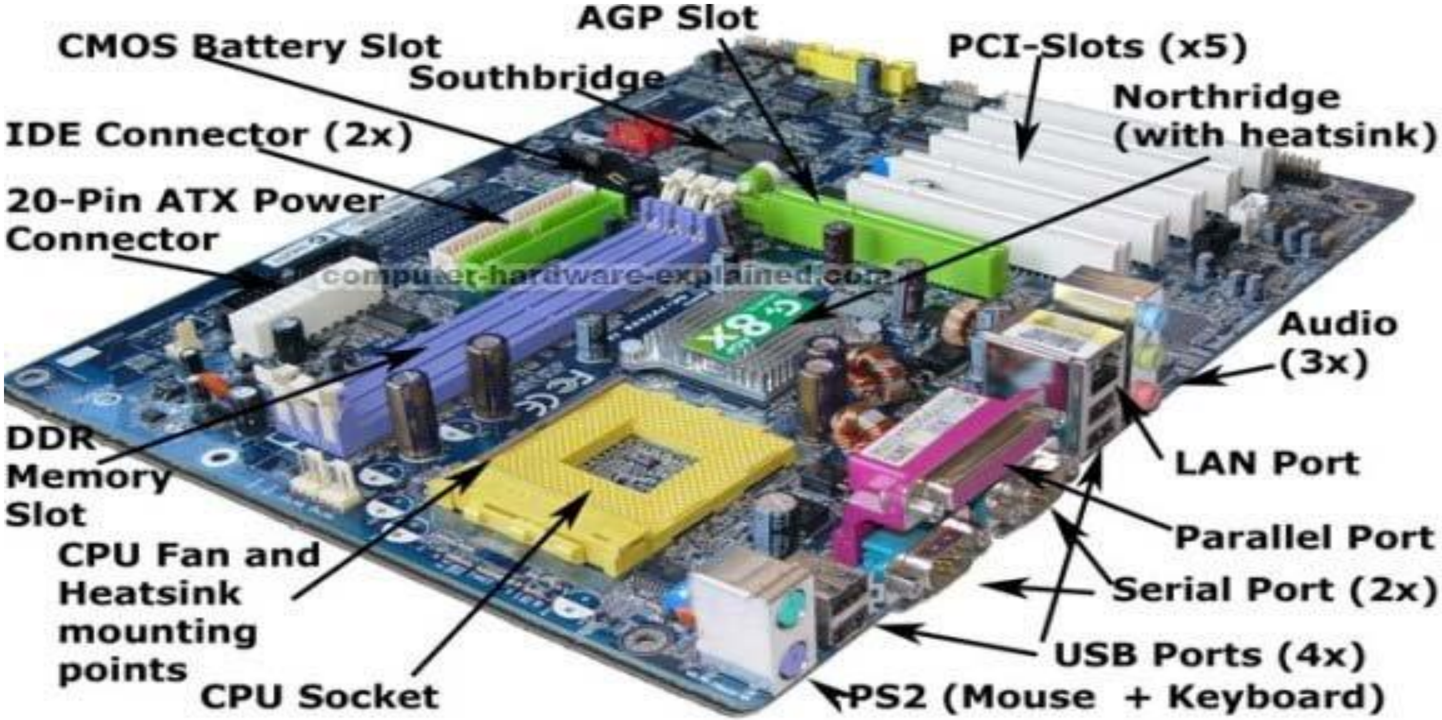
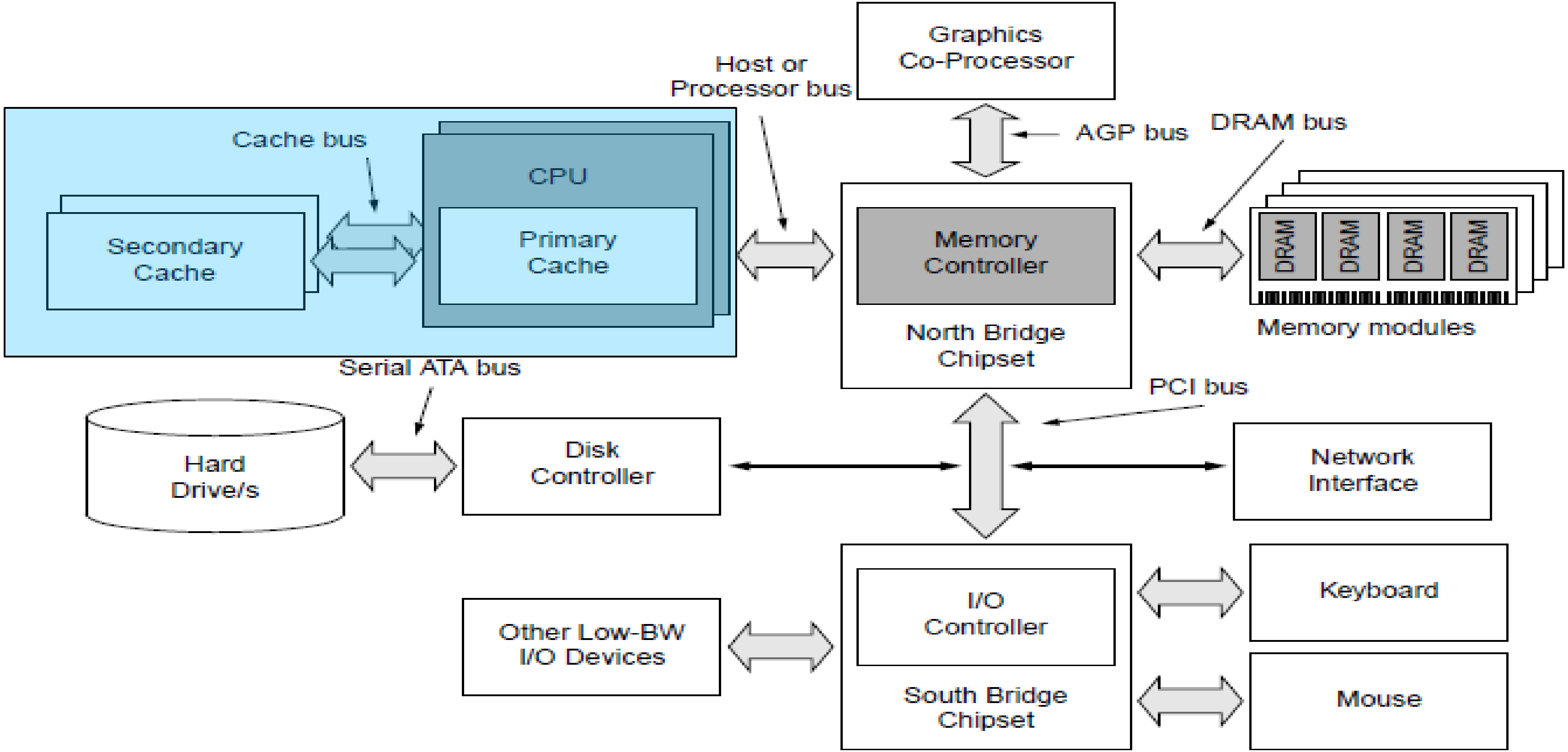
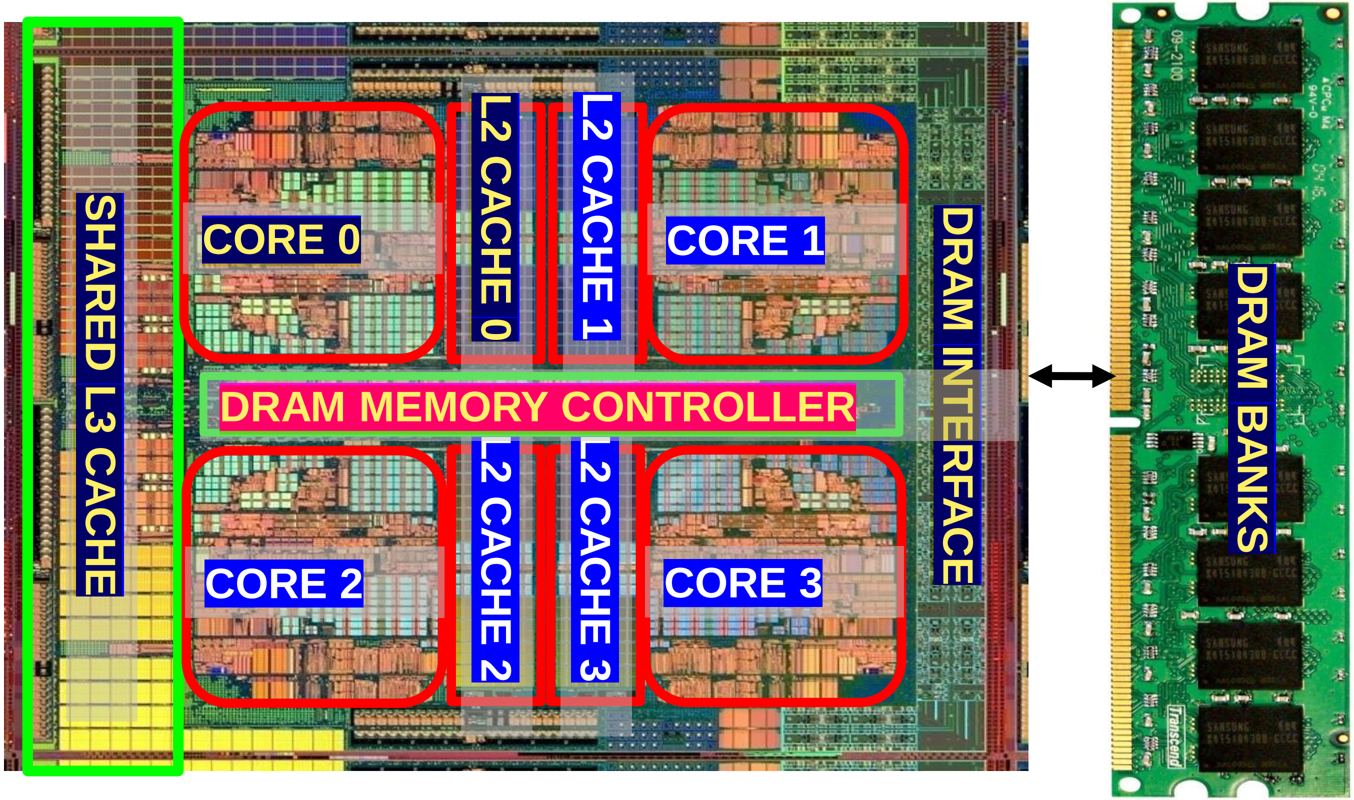
Components of a Modern Computer



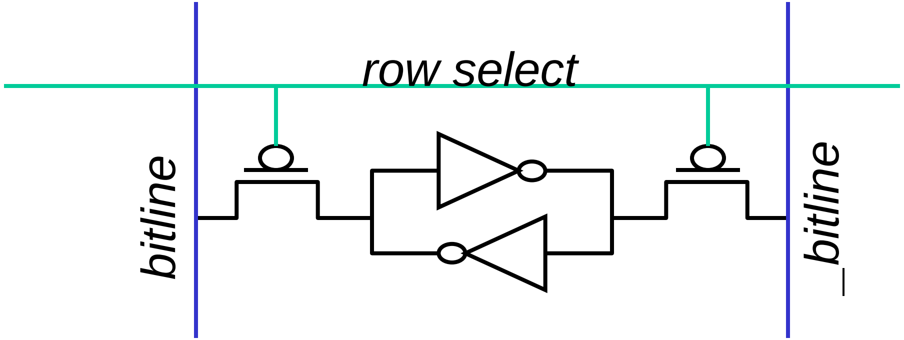
Components of a Modern Computer



Main Memory in the System

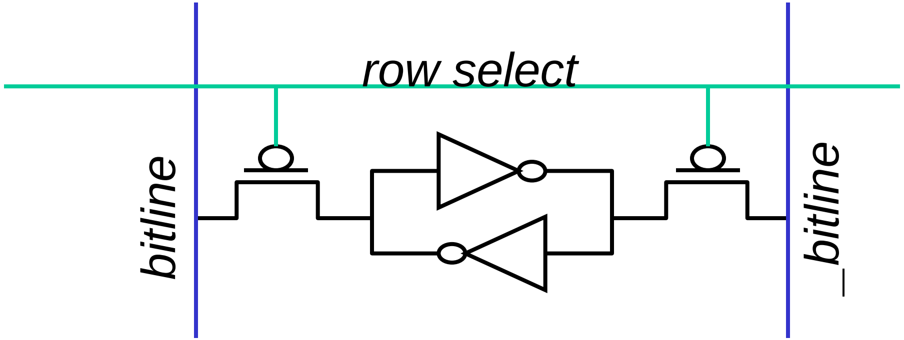


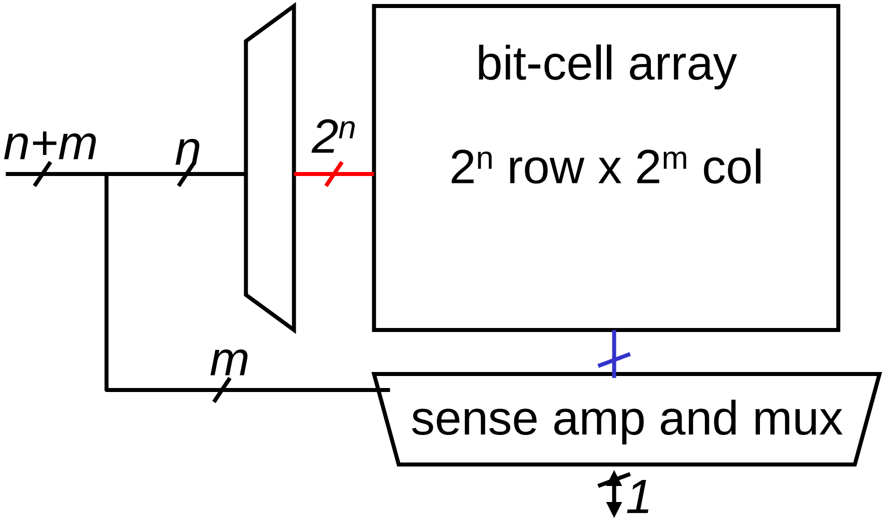
SRAM (Static Random Access Memory)



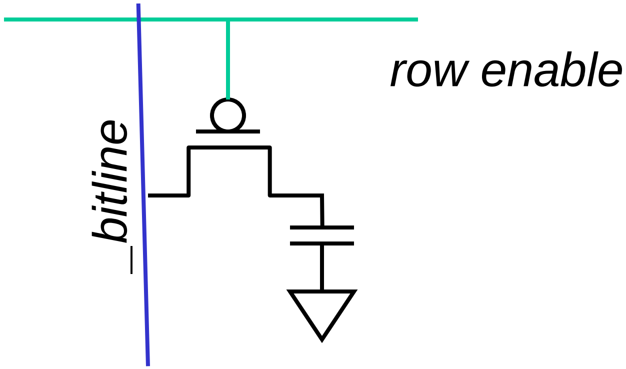
- **Read Sequence**
- address decode
- drive row select
- selected bit-cells drive bitlines (entire row is read together)
- Col. Select (data is ready)

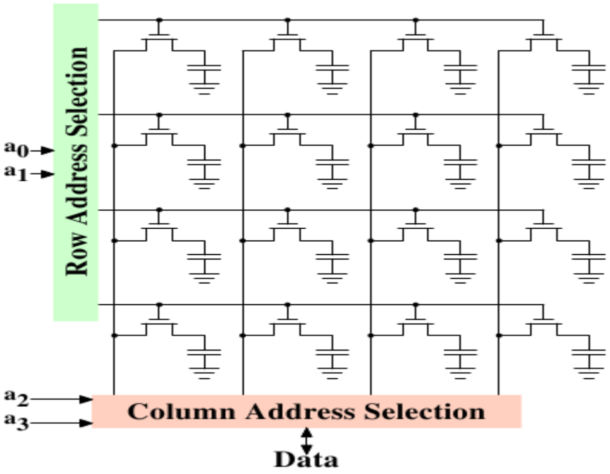
SRAM (Static Random Access Memory)





DRAM (Dynamic Random Access Memory)





- Bits are stored as charges on capacitor
- B cell loses charge when read
- B cell loses charge over time
- A"flip-flopping" sense amplifier amplifies and regenerates the bitline, data bit is mux'ed out

DRAM vs SRAM

- **DRAM**
  - Slower access (capacitor)
  - Higher density (1T, 1C cell), Lower cost
  - Requires refresh (power, performance, circuitry)
  - Manufacturing requires putting capacitor and logic together
- **SRAM**
  - Faster access (no capacitor)
  - Lower density (6T cell), Higher cost
  - No need for refresh
  - Manufacturing compatible with logic process (no capacitor)

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